

Butler University

Las Chicas Superpoderosas y David

Insurance Claim Management System

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Abstract

This report presents the design and development of an Insurance Claim Management System created as part of a software engineering project at Butler University. The system was designed to streamline the insurance claims process by providing an intuitive graphical user interface for multiple user roles, including clients, claim managers, finance managers, and administrators. Core functionalities include claim submission, status tracking, communication between users, and administrative oversight of accounts and system data.

The project was developed using Agile methodology, specifically the Scrum framework, across six iterative sprints. Each sprint focused on incremental improvements such as refining user stories, building and integrating the database, developing user interfaces, and implementing key system functionalities. Throughout the development process, the team encountered challenges related to version control, time management, and technical integration, which were addressed through improved communication, branching strategies, and collaborative problem-solving.

In addition to technical implementation, the project emphasized object-oriented design principles and a three-tier architecture to enhance system organization, scalability, and maintainability. The experience provided valuable insights into Agile development practices, teamwork, and real-world software engineering workflows. Ultimately, the team successfully delivered a functional application and gained practical knowledge that will inform future software development endeavors.

Chapter 1: Introduction

Problem Statement and Requirements

Our group decided to do the Insurance Claim Management System Project, which required us to develop an application for clients to apply claims that are then sent to members of an insurance company, like a claim and finance manager, to evaluate their claim accordingly. The claim management system provides a clear and easy to use GUI interface for all users involved (Client, Claim Manager, Finance Manager, and Admin). As mentioned earlier, the client is able to apply and then view the status of their claims, and also send messages to anyone who they deem necessary. The Claim Manager evaluates claims that the clients apply and determine whether or not they are eligible for compensation. From there, the Claim Manager can send claims over to the Finance Manager who is in charge of giving the clients proper compensation for their accidents if they are eligible for it. The Admins for this management system are in charge of keeping the system clean and up to date. They are in charge of approving and denying new accounts, managing different users' accounts in the system and overall just making sure that the application runs smoothly. All users in the system also have the ability to edit their accounts and send messages to whomever that has a valid account.

Team Structure

For our team, David Jefferson was the Scrum Master of the group and was in charge of controlling the traffic and facilitating the use of SCRUM to complete sprint goals. Katie Baumgartner was the Full Stack and oversaw working on connecting our frontend and backend aspects of the project together. Lizy Argueta was our Database Backend manager and oversaw our database and backend features of the project. Braydon Sheets was our GUI Frontend

developer and focused on designing the UI/UX for the project. Lastly, Sophie Resner also had Full Stack duties as she also worked on both frontend and backend of our project.

Organization of the Report

This report is organized into three main sections. The first section covers the sprints, including details about our meetings, sprint demos, the tools we used, and our reflections on each sprint cycle. The second section presents the conclusion and lessons learned, focusing on our experience with Scrum methodology and the key accomplishments achieved throughout the project. Finally, the third section outlines the team organization and roles, explaining who was responsible for each part of the project and how each member contributed to bringing the final product to life.

Chapter 2: Project Sprints

Sprint 0

Sprint Planning, Meeting, Goals

For Sprint 0, we created teams, selected our project and set up all the tools we needed. During our first meeting, we created goals and picked the Insurance Claim Management System as the application we wanted to develop. Our main goals for this Sprint were to assign team roles, pick our project, set up all the software and tools needed and then create user stories and set up our initial ERD.

Product and Sprint Backlog

We set up Visual Studio, created an iMessage group chat, created our group's Jira board, assigned roles, picked a project, logged user stories and created an ERD. (See Appendix C)

Daily StandUps

We had two standup meetings for this sprint along with an outside class meeting in which we discussed getting comfortable and familiar with the software along with sharing our progress creating user stories.

Tools Used

We used iMessage and Jira for this sprint.

Sprint Demo and Accomplishments

For this sprint, we completed all of our goals we set for ourselves as a team. We're still working on our User Stories and our ERD, while still getting comfortable with the software.

Sprint Retrospective

For this sprint we did not encounter too many issues. Our biggest challenge was time but as far as working in a group we had great communication amongst each other. One other

challenge we faced was understanding all the new terminology so we plan to figure out these new concepts in order to make our lives and projects easier.

Sprint 1

Sprint Planning, Meeting, Goals

For Sprint 1, we set out to improve our User Stories and ERD since we received feedback from our last sprint to improve both our User Stories and ERD. We also set out to begin work on the project by creating a database so that users could login and create an account. For this sprint we also made it a goal to begin working on design or UI forms. So all in all, we needed to create a user interface, a database in order to keep track of users and connect the two pieces.

Product and Sprint Backlog

We used our login user stories for this sprint (Create + design login and registration user forms). We added database tables to our ERD, added basic functionalities and made a landing page for all the users. (See Appendix C)

Daily StandUps

We had two standup meetings for this sprint along with two outside class meetings on Monday and Wednesday in which we discussed our progress on our goals. Lizy Argueta shared her progress on creating the ERD and database for our application. Braydon Sheets discussed his progress creating different UI forms. Katie Baumgartner and Sophie Resner stated their progress connecting our database to our login and registration forms.

Tools Used

For this sprint we used Visual Studio, AzureLabs, GitHub, Jira and iMessage.

Sprint Demo and Accomplishments

For this sprint, we completed most of our goals we set for ourselves as a team. This sprint we ran into problems with our GitHub along with merging issues. We had major problems trying to merge and sync different features all onto the same branch so we couldn't quite finish all the goals we had set out for the team.

Sprint Retrospective

Most of our issues for this sprint came from GitHub and couldn't finish all our goals. We want to continue to remain resilient with our problem-solving but we also want to be better at working with Visual Studio and GitHub errors so they don't drastically affect us later on in our project.

Sprint 2

Sprint Planning, Meeting, Goals

For Sprint 2, we had to take some extra time in order to clean up some of the mistakes we had in the last sprint. Therefore, our goals for Sprint 2 were to clean up our GitHub code so we can merge code together, then refine our login and registration pages, after that reconnect our login and registration UI forms to our database, and then create home landing pages for different users.

Product and Sprint Backlog

We used our admin and login user stories for this sprint. We connected our backend to our frontend, updated our account creation form, created other user forms, cleaned up GitHub repository, connected registration to user database, and connected the client form to the login form. (See Appendix C)

Daily StandUps

We had two standup meetings for this sprint along with two outside class meetings on Monday and Wednesday in which we discussed our progress on our goals. Lizy Argueta shared her progress on updating our ERD and database for our application. Braydon Sheets discussed his progress creating more UI forms. Katie Baumgartner and Sophie Resner stated their progress linking our database and our frontend.

Tools Used

For this sprint we used Visual Studio, AzureLabs, GitHub, Jira and iMessage.

Sprint Demo and Accomplishments

For this sprint, we ran into many troubles again with GitHub but regardless, we were still able to complete the goals we set out for the team. We had another problem with GitHub and merging issues so we had to do a lot more work just to have a working demo for this sprint.

Sprint Retrospective

Although we were able to complete all our goals, we had to do a lot of unnecessary work just to have a working demo. As a solution to our problem, we decided to have each member create their own branch in GitHub that they work on themselves to then sync and merge to whomever they need too. We decided that we we're going to be better at communicating different issues we were having rather than trying to solve them ourselves. The only thing we decided to keep up with is our resilience because no matter what happened, we continued to grind and come up with something to present.

Sprint 3

Sprint Planning, Meeting, Goals

For Sprint 3, we set out to implement some of the User Stories we had come up with for the clients that use our Insurance Claim Management System (Apply Claim form + Profile

window used to edit profiles). At the same time we also had to fix a small database issue we were having in Visual Studio in order to progress further in our project.

Product and Sprint Backlog

We used our client user stories for this sprint (ApplyClaim + edit profile window). We fixed our database in Visual Studio. (See Appendix C)

Daily StandUps

We had two standup meetings for this sprint along with two outside class meetings on Monday and Wednesday in which we discussed our progress on our goals. Lizy Argueta shared her progress on fixing our database problem. Braydon Sheets shared his progress creating different buttons for our client page. Katie Baumgartner shared her progress connecting our database to our login and registration forms and Sophie Resner provided insight as to how she was supporting Lizy and Braydon on both their ends along with the creation of our UML Class Diagram.

Tools Used

For this sprint we used Visual Studio, AzureLabs, GitHub, Jira and iMessage.

Sprint Demo and Accomplishments

For this sprint, we completed the majority of our goals we set for ourselves as a team. This sprint we ran into problems with time and other class conflicts. Since we didn't have that much to complete our goals, we emphasized the things we could show for our live demo like the user functionalities.

Sprint Retrospective

Since we all had the same class conflict along with time issues, we decided and agreed on as a team to implement what we can show and then next sprint to give it more attention. We did

good at communicating the problems we were having and how we came together to come up with a solution to get through the problem we were having.

Sprint 4

Sprint Planning, Meeting, Goals

For Sprint 4, we set out to implement the universal features that all users have like the messaging feature and the edit profile feature. Since we were not able to get as much done as we wanted to last sprint, we took it upon ourselves to overload just to stay on track and continue progressing well on our project. We also had to fix our ApplyClaim function for our clients since it was not working all the way properly from the last sprint. Lastly, we had to update our UML Class Diagram since we had received constructive feedback on our diagram.

Product and Sprint Backlog

For this sprint we were focused on the User Stories that we found to be applicable to all users such as the SendMessage and Edit Profile. (See Appendix C)

Daily StandUps

We had two standup meetings for this sprint along with four outside class meetings on Mondays and Wednesdays in which we discussed our progress on our goals. Lizy Argueta shared her progress adapting our database to handle messages and the updated profile information for users. Braydon Sheets shared his progress creating different UI forms for more users. Katie Baumgartner shared her progress connecting our database to our login and registration forms and Sophie Resner added how she helped create UI forms and connect these made forms to our database.

Tools Used

For this sprint we used Visual Studio, AzureLabs, GitHub, Jira and iMessage.

Sprint Demo and Accomplishments

For this sprint, we completed the majority of our goals we set for ourselves as a team. We set out a lot of goals to complete in a short amount of time along with having to handle work from other classes. For the goals we left unfinished we had made good progress on them in order to wrap up our project for Sprint 5.

Sprint Retrospective

Although we're happy about what we've accomplished for this sprint, we knew we'd have to really come together for this last sprint and go all hands on deck in order to complete the specified requirements by the product owner.

Sprint 5

Sprint Planning, Meeting, Goals

For Sprint 5, our goals were pretty clear. Implement everything we hadn't already. We had a couple of user functionalities that we hadn't gotten to yet. These tasks included implementing small features for our Admin like viewing users, Claim Manager viewing claims, and our Finance Manager handling the financial aspect of a claim.

Product and Sprint Backlog

For this sprint we were focused on the User Stories that we hadn't gotten to yet. (See Appendix C)

Daily StandUps

We had two standup meetings in class for this sprint along with a meeting everyday where we went over what we were currently working on and how we could help one another out. Lizy Argueta shared her progress about the finishing touches she put onto our database so that all

our users, claims and messages worked for our application. Braydon Sheets shared his progress finishing the final UI pages he made. Katie Baumgartner shared her progress connecting everything we had together and adding all the functionality we needed. Sophie Resner shared how she helped add some of the functionalities we needed for some users along with the creation of a few UI forms.

Tools Used

For this sprint we used Visual Studio, AzureLabs, GitHub, Jira and iMessage.

Sprint Demo and Accomplishments

For this last sprint, we completed all of our tasks and were able to present a working demo for the class.

Sprint Retrospective

We were extremely proud of what we accomplished and with how we presented our final work to everyone, even though we were completely gassed and barely staying awake.

Chapter 3: Conclusion and Lessons Learned

At the beginning of the semester, we had little to no understanding of what agile development was. However, this project certainly gives us first-hand experience with using this type of structure and technique. Working in these iterative, timed sprints gave us more time and leeway to finish small achievable goals, instead of getting overwhelmed with the project. We had been so used to you personally having to figure out how you were going to do things and not always given much of an actual schedule, this approach made the project seem slightly more manageable. It showed that when it comes to software engineering, this iterative and agile development can be very helpful, with the importance of communication and teamwork.

Using an object-oriented approach was very crucial to our project. Using and designing things such as a 3-tiered architecture made our system a lot clearer and more manageable. Boundary/Entity/Controller classes helped reduce the complexity of individual aspects. From our experience, we realized how valuable an object-oriented approach was, and that behind a good design is intentional planning and frequent revisiting, through things such as retrospective sprints.

Working as a team taught us a ton about flexibility, communication, and collaboration. One of the biggest challenges that we face is having time for all of us to work together and also continuing to have a sense of teamwork even though times got tough and mistakes were being made. The times we were usually all available were during our scheduled class, sometimes even a little bit afterwards, and late into the night, a few days during the semester. We had a lot of team challenges with some members getting distraught or overwhelmed at times. Even though emotions ran high throughout the squad, nobody gave up and everyone was able to push through all the obstacles in our way in order for us to keep going to finish our project. With our successes

and challenges, we gained valuable experience in teamwork that would most definitely be useful in the future.

Chapter 4: Team Organization and Roles

The Scrum leader, David Jefferson, facilitated the team's use of the SCRUM Framework to achieve the team's goals. The Scrum Leader led the daily standups, sprint retrospectives, and sprint demonstrations. He oversaw the sprint goals and kept track of the group's progress. Occasionally, David would step in to help the team members with coding and testing. Each and every one of the team members participated in the creation of team goals, the sprint demos, and the sprint retrospectives. Our Full Stacks, Katie Baumgartner and Sophie Resner, oversaw working on both the frontend and backend of the project. They often worked more on the frontend since that was what the group was mainly focused on making and improving. Our Database Backend Developer, Lizy Argueta, oversaw the database creation and backend code for the project. The backend developer created the ERD Diagram and our database tables for the project. The Frontend Developer, Braydon Sheets, oversaw the user interface and some of the design of our project. The Frontend Developer designed a few of the forms. The tasks for each sprint were divided between the members based on frontend and backend, so each member of the development team worked on their specific tasks. Each member of the team efficiently and effectively carried out their tasks and contributed to the final project.

Appendices

Appendix A **Sprint Presentations**

Sprint 0 Demo presentation. This was the first presentation we have and we discussed an introduction to our project and SCRUM principles. We also introduced user stories in this sprint.

[Sprint 0 Presentation](#)

Sprint 1 Demo Presentation. This was our presentation where we discussed our goals for the project, progress and difficulties. We also did our first live demo of our project. [Sprint 1](#)

[Presentation](#)

Sprint 2 Demo Presentation. In this presentation we discussed our improved user stories, goals, and candidly explained the issues we ran into during this sprint. [Sprint 2 Presentation](#)

Sprint 3 Demo Presentation. In this presentation we first introduced the idea of a UML class diagram, updated our goals, along with our user stories. We also added a flow chart to be able to show the inner workings of our project and how everything is connected. [Sprint 3 Presentation](#)

Sprint 4 Demo Presentation. In this presentation we showed an improved UML class diagram using visual studio, we also indicated with colors which classes have been implemented and which still need to be, and provided an updated ERD. [Sprint 4 Presentation](#)

Sprint 5 Demo Presentation. In the final presentation we showed a completed UML class diagram, along with a completed project. We were able to show that a lot of progress was made between sprint 4 and sprint 5. [Sprint 5 Presentation](#)

Appendix B

Peer Evaluations

Sprint 0 Feedback: For our sprint 0 we got mostly very positive feedback from our peers, the biggest improvement was to improve our ERD, and to start making a design for our GUI. Along with starting to implement some of the classes from our ERD into the project.

Sprint 1 Feedback: For our sprint 1 we again got mostly positive feedback from our peers, the biggest improvement we were told was that we did not properly explain what our objective of the project was. Along with making sure to use “we” instead of “I”, and trying to make/create a design for our GUI.

Sprint 2 Feedback: For our sprint 2 we did not get as much positive feedback as we had seen in previous sprints. We had discussed that there were a lot of mistakes that lead us to not be able to present a bunch of information for this sprint. The biggest area of improvement we were told was to fix our problems for the future so that we do not run into this issue again, to have more energy , and speak louder.

Sprint 3 Feedback: For our sprint 3 we got good feedback. What we took away from this was to fix the design of our GUI, the words were hard to read based on the color, and to be a little more enthusiastic. We had been told this before along with group members speaking up, we took this very seriously as it had been told to us a couple times.

Sprint 4 Feedback: In our sprint 4 we got good positive feedback. What we mostly took away from this presentation was the completion of our UML class diagram and ERD along with the functionality of our project along with refactoring. These were very strong points that we took very seriously into consideration before the final sprint.

Sprint 5 Feedback: For our final sprint we were not given any feedback at the time of writing this report.

Appendix C

Project Diagrams

User Stories:

The user stories were made by all of the team members including David Jefferson, Katie Baumgartner, Lizzy Argueta, Braydon Sheets, and Sophie Resner.

Client User Stories:

- **Login 1 / 5 Difficulty**
As a client using this claim management system.
I want to be able to login to the system.
So that I can file claims, view my claim status, edit my profile and post comments.
- **Apply Claim 3 / 5 Difficulty**
As a client using this claim management system.
I want to be able to apply for insurance claims.
So that I can request compensation for an incident.
- **View Claim Status 2 / 5 Difficulty**
As a client using this claim management system.
I want to be able to view the status of a claim that I've filed.
So I can see how far along the claim is and act accordingly.
- **Edit Profile 2 / 5 Difficulty**
As a client using this claim management system.
I want to be able to edit my profile.
So I can update any information about me if anything needs to be.
- **Post Comment 2 - 3 / 5 Difficulty**
As a client using this claim management system.
I want to be able to post comments on an insurance claim.
So I can respond to other parties if need be.
- **Logout 1 / 5 Difficulty**
As a client using this claim management system.
I want to be able to logout of the system.
So I can stop using the system after I am done with it.
- **Communication 3-4 / 5 Difficulty**

As a client using this claim management system.

I want to be able to communicate with an agent to understand the status and direction of where my claim is heading.

- **Security 3-4 / 5 Difficulty**

As a client using this claim management system.

I want to be able to have a secure login to the management system,

Using something along the lines of two-factor authentication.

Claim Manager User Stories:

- **Login 1 / 5 Difficulty**

As an Insurance Claim manager.

I want to be able to login to the claim management system.

So I can manage claims.

- **View Claim Status 2 / 5 Difficulty**

As a Claim Manager using this claim management system.

I want to be able to view the status of a claim that has been filed.

So I can see what the status of the claim is and act accordingly.

- **Edit Profile 2 / 5 Difficulty**

As a claim manager.

I want to be able to edit my profile.

So I can make changes to it if needed.

- **Logout 1 / 5 Difficulty**

As a Client Manager.

I want to be able to logout of the system.

So I can stop using the system when I am done.

- **Evaluate Forms Difficulty 2 / 5**

As a Client Manager

I want to be able to evaluate forms submitted by clients

So I can keep the insurance claim process moving forward smoothly

- **Update Status Difficulty 1 / 5**

As a Client Manager

I need to be able to update the status of a claim

So I can keep the process as up to date as possible

- **Generate Reports Difficulty 3 / 5**

As a Client Manager

I need to be able to generate reports

So I can write up reports if needed during the process

Admin User Stories:

- **Login 1 / 5 Difficulty**

As an Admin of the claim management system.

I want to be able to Login to the system.

So I can oversee the system and make sure everything is working properly.

- **Register CliMan (Claimant) 3 - 4 / 5 Difficulty**

As an Admin of the claim management system.

I want to be able to register a claimant to a claim.

So the claim process can proceed smoothly with the proper individuals.

- **Register FinMan (Finance Manager) 3 - 4 / 5 Difficulty**

As an admin of the claim management system.

I want to be able to register a Finance Manager to a claim.

So they can oversee the claim and manage its finances properly.

- **Logout 1 / 5 Difficulty**

As an Admin

I want to be able to logout of the system.

So I can stop using the system when I am done.

- **Apply Claim Difficulty 3 / 5**

As an Admin

I want to be able to apply claims

So the correct clients correspond to the correct claims

- **View Claim Status Difficulty 1 / 5**

As an Admin

I want to be able to view the status of claims

So I can see how far along a claim is

- **Generate Reports Difficulty 3 / 5**

As an Admin

I want to be able to generate reports

So I can create reports for each claim

- **Employee code Difficulty 2 / 5**

As an Admin

I want to be able to have a code unique to Admins, to increase security and so that only people who are admins can view private client information.

- **Generate ID Difficulty 2 / 5**
As an Admin
I want to be able to generate ID's
So each claim and user can have a unique and identifiable ID
- **Maintain DB Difficulty 2 - 3 / 5**
As an Admin
I want to be able to maintain the database
So I can keep all of the data in the system up to date and in order
- **Validate Unique ID Difficulty 2 - 3 / 5**
As an Admin
I want to be able to Validate Unique ID's
So I can make sure that an ID corresponds to an actual claim or user
- **View user details on a particular date Difficulty 2 / 5**
As an Admin
I want to be able to view user details on a particular date
So I can see information about a user at a certain time to see if or how things have changed
- **Transfer amount details Difficulty 2 - 3 / 5**
As an Admin
I want to be able to Transfer Amount Details
So I can keep the details of a certain claim or user attached to that particular claim or user

Finance Manager User Stories:

- **Login 1 / 5 Difficulty**
As a Finance Manager.
I want to be able to login to the claim management system.
So I can work on the cases that I am involved in.

- **View Claim Status 2 / 5 Difficulty**

As a Finance Manager.

I want to be able to view the status of claims that clients have filed.

So I can manage the related finances accordingly.

- **Update Status 1 - 2 / 5 Difficulty**

As a Finance Manager.

I want to be able to update the status of claims.

So I can make sure everything having to do with the claim is up to date.

- **Logout 1 / 5 Difficulty**

As a Finance Manager.

I want to be able to logout of the system.

So I can stop using the system when I am done.

- **Evaluate amount details Difficulty 2 / 5**

As a Finance Manager

I need to be able to evaluate amount details

So I can keep track of our company's finances

- **Security Measure Difficulty 3 - 4 / 5**

As a Finance Manager

I need to be able to have a secure way to store a claim after it is created.

- **Transfer Amount Details Difficulty 2 - 3 / 5**

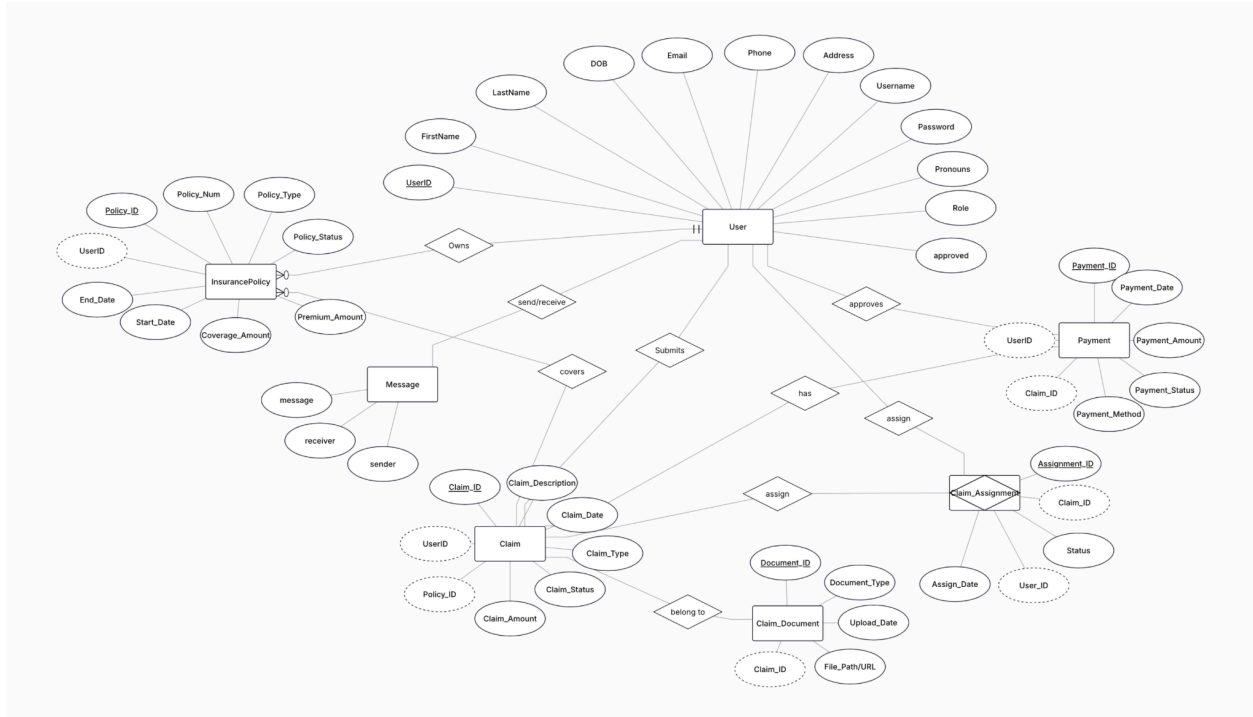
As a Finance Manager

I need to be able to transfer amount details

So I can keep appropriate parties up to date

ERD:

The ERD was made by Lizzy Argueta using ERD plus.



UML Class Diagram:

This class diagram was made by Sophie Resner, using visual studio.

